Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Full-Stack Developer with **3 years** of industry experience building and supporting internal web applications for **investment** teams at an asset manager: **ReactJS** frontends, **Python FastAPI** backends and a **PostgreSQL** data platform, released to real users and improved on their feedback. Strong **SQL** and **Python**, comfortable diagnosing issues in live systems and turning requirements from **investment**, Compliance and Client Service stakeholders into working tools; **MSc in Data Science at the University of Bristol** (awaiting result). Seeking a **Full-Stack Developer** role in **front-office investment technology**, bringing relational database experience from **PostgreSQL** to **SQL Server**.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Full-Stack Development: React, Python & SQL

- Designed and built internal web applications end to end for investment, Compliance and Client Service users: **ReactJS** frontend served by **Nginx**, **Python FastAPI** microservices in **Docker**, and **Azure OpenAI** features (RAG on **pgvector**, tool-calling agents running **Python** functions over **SQL** data).
- Developed **Power BI** dashboards for the investment team's fortnightly credit report analysis, improving analyst efficiency by **10 hours a week**, then recreated them in the **ReactJS** frontend so the whole organization could use the data at lower cost than Power BI licences.
- Designed the entire **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** ingestion scripts for the **Morningstar API**, covering every fund in the Indian mutual fund market (2023–2025), **integrated** with the **legacy data** in analysts' Excel files; developed **Optimized SQL** queries, **SQL Functions** and **Procedures**.

Live Support, Production Issues & Stakeholder Delivery

- Released each application to real users and changed it based on what they found; built a **WhatsApp** fund-comparison bot (**Azure OpenAI** tool calling) used daily by the Sales team (roughly **50 to 100 users**, **400 to 1,000 requests** in working hours), and fixed tool-call failures from poorly extracted parameters by adding example inputs to the JSON tool definitions.
- Built a **Python** job that parses over **300 daily emails** from credit rating agencies and sends analysts a daily summary, saving **60 manual analyst hours a month**; fixed the parser when one agency switched to image emails by replacing **AWS Textract** with **Azure OpenAI**, which also improved accuracy.
- Analyzed data with **Python Pandas** and identified buying patterns using **Market Basket Analysis** (**Apriori Algorithm**), handing the insights to the marketing team.
- Worked directly with **Investment, Compliance and Client Service** stakeholders, with Compliance reviewing every application before release; **de facto team lead for up to 4 interns**; **Best Team Project** twice (2024 rating-email automation, 2025 department RAG chatbot).

Cloud Infrastructure, Security & Deployment

- Designed the infrastructure architecture on **AWS** (**EC2, S3, ECR, Lambda**) and automated deployment of separate frontend and backend repositories (dev and prod) through a one-click **Jenkins and GitLab CI/CD** pipeline.
- Implemented **RBAC** with **LDAP** authentication, restricted access to company private IPs with **AWS Security Groups**, and deployed an **Nginx** reverse proxy as one of the fixes that took the application through the security team's **VAPT**.

PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built with **FastAPI** services and a **React TypeScript** frontend. My focus: the **data preprocessing pipeline** (17 raw inputs into 14 clean, joined tables, with integrity checks and a build manifest audit file), the **scoring and ranking algorithm** and the **RAG-based AI assistant**. Refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **Wingman:** AI dating-coach web app with 108 LLM personas across 12 archetypes (**React** with Vite and Tailwind, **Python FastAPI**). Streams results to the UI over server-sent events, validates LLM output with **Pydantic** and caches results by input hash. GitHub: https://github.com/Arishtanemi4/wingman.git

ACHIEVEMENTS

- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle credit card fraud detection problem.)
- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework for AI-assisted coding) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git

SKILLS

- **Tools:** Jupyter, Anaconda, AWS (EC2, S3, ECR, Lambda, SageMaker, EventBridge, Textract), Azure OpenAI, Power BI, Docker, Jenkins, GitLab, Git, Nginx, Linux, LDAP, VS Code, Claude Code.
- **Languages:** Python (pandas, FastAPI, scikit-learn), ReactJS, SQL, NodeJS, R (dplyr, tidyverse, ggplot2), Shell.
- **Databases:** PostgreSQL (normalized schema design, SQL functions and procedures, query optimization, pgvector), SQLite.